Two microRNA axes, from measured kinetics to a computed outcome

Every number below is loaded from results/tables/ and data/extracted/ at draw time - none is retyped from a sentence

ALZHEIMER'S DISEASE AXIS

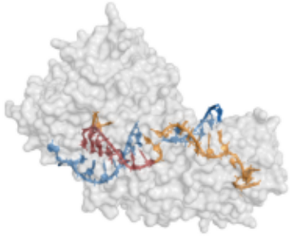
miR-29b / miR-29c
t½ 7 h / 10.6 h
not one species [K017,K018]

-| BACE1
BACE1 : APP ≈ 1 : 54
(116 vs 6284 copies/bouton) [K055,K056]

Aβ42 production
6.7 (ctrl) / 6.6 (AD) %/h, n.s.
clearance −30.3% in AD [K001,K002]

AD / control Aβ ratio
= 1.41×
no free parameter

Required BACE1 knockdown
to close the gap: 29%
(< 50% measured, Hebert 2008)



AGO2 loaded with a guide RNA
(PDB 6N4O, human AGO2-miR-122)
the seed-pairing step common to both axes

Mimic dosing penalty: 7.7× peak
vs a median-stability miRNA —
needs ~20× stabilisation

miR-7
t½ 1.7 h — shortest surveyed
(median 34 h) [K020,K016]

-| SNCA / α-synuclein
21.6 µM in the presynaptic
bouton [K054, Wilhelm 2014]

Elongation regime
30–32% of V_max
below half-sat. (m½ 46–50 µM)

Measured knockdown 30% / 43% (Doxakis 2010)
→ 23–34% slower elongation

PARKINSON'S DISEASE AXIS

Sources: Mawuenyega et al. 2010; Hébert et al. 2008; Zhang et al. 2011; Kingston & Bartel 2019; Wilhelm et al. 2014; Buell et al. 2014; Doxakis 2010; Tushev et al. 2018 - full derivation in docs/en/METHODS.md section 11.